Tutorial 4
Reliable Data Transfer and Go-Back-N

Instructions:

Unless stated otherwise, assume the standard Go-Back-N (GBN) protocol. The receiver uses cumulative acknowledgements, accepts only the next expected packet, and discards out-of-order packets. An ACK carrying sequence number k indicates that all packets up to $k-1$ have been received correctly and that packet k is the next expected packet.

Q1. Go-Back-N: Cumulative ACKs and Retransmission

A Go-Back-N protocol uses a 4-bit sequence number field and a sender window size of 7. The sender transmits packets with sequence numbers

$$3, 4, 5, 6, 7, 8, 9.$$

The receiver initially expects packet 3. Packet 5 is lost during transmission, while packets $3, 4, 6, 7, 8$, and 9 reach the receiver successfully. The receiver follows the standard GBN rule of discarding out-of-order packets and sending cumulative ACKs.

(a) What ACK does the receiver generate after receiving packet 3?

(b) What ACK does the receiver generate after receiving packet 4?

(c) What ACK does the receiver generate when packets 6, 7, and 8 arrive? Explain whether the ACK changes after each arrival.

(d) The sender's timer for the oldest unacknowledged packet expires. Which packet(s) must the sender retransmit?

(e) Suppose packets $5, 6, 7, 8$, and 9 are successfully retransmitted. What cumulative ACK should the receiver generate after receiving the complete sequence?

Q2. Go-Back-N: Multiple Packet Losses

A Go-Back-N protocol uses a sender window size of 5. At time $t = 0$, the sender transmits packets with sequence numbers

$$10, 11, 12, 13, 14.$$

Packets 11 and 13 are lost. Packets 10, 12, and 14 reach the receiver successfully. The receiver initially expects packet 10 and discards all out-of-order packets.

(a) Determine the ACK generated after packet 10 is received.

(b) Determine the ACK generated when packet 12 arrives.

(c) Determine the ACK generated when packet 14 arrives.

(d) Identify the oldest unacknowledged packet at the sender.

(e) The sender's timer for the oldest unacknowledged packet expires. Which packets must be retransmitted?

(f) Suppose packet 11 is successfully retransmitted, but packet 13 is lost again. Describe the ACK behavior. Which packet remains the oldest unacknowledged packet at the sender, and which packets will be retransmitted when its timer eventually expires?

(g) Determine the minimum total number of packet transmissions required for packets 10–14 to be successfully delivered under this loss pattern, assuming no additional losses.

Q3. Go-Back-N: Sequence Number Wrap-Around

A Go-Back-N protocol uses a 3-bit sequence number field. The sender window size is 5. The receiver initially expects sequence number 5.

The sender transmits the following packets:

$$5, 6, 7, 0, 1.$$

Packet 7 is lost, while packets 5, 6, 0, and 1 reach the receiver in that order before packet 7 is retransmitted.

The receiver follows the standard GBN rule: it accepts only the next expected packet, discards out-of-order packets, and sends cumulative ACKs.

(a) What ACK does the receiver generate after receiving packet 5?

(b) What ACK does the receiver generate after receiving packet 6?

(c) What happens when packet 0 arrives while the receiver is still expecting packet 7?

(d) What happens when packet 1 subsequently arrives?

(e) Explain why sequence-number wrap-around can create ambiguity if the sender window is chosen incorrectly.

(f) For an m-bit sequence number field, determine the maximum sender window size that guarantees unambiguous operation in standard Go-Back-N.

(g) For $m = 3$, explain what ambiguity could arise if the sender window size were 8.

Q4. Reliable Data Transfer: Designing an RDT Protocol

Consider a communication channel in which packets and ACKs may be corrupted, lost, or delayed. Packets may also be duplicated, but the channel does not reorder packets. You are asked to design a reliable data transfer protocol using:

- checksum,
- sequence numbers,
- acknowledgements,
- retransmission timers.

(a) Explain why a checksum alone cannot guarantee reliable delivery.

(b) Explain why sequence numbers are required when ACK packets may be corrupted, lost, or delayed.

(c) Design the sender and receiver state transitions for a stop-and-wait RDT protocol. Clearly specify the actions taken when a DATA packet or ACK is corrupted.

(d) Consider the following event sequence:

$$\text{DATA}_0 \rightarrow \text{ACK}_0 \rightarrow \text{ACK}_0 \text{ is lost} \rightarrow \text{DATA}_0 \text{ is retransmitted.}$$

What should the receiver do when the duplicate DATA_0 arrives?

(e) Explain why the receiver must not deliver the duplicate DATA_0 to the application layer.

(f) Explain how sequence numbers and receiver state together prevent duplicate application-layer delivery in this situation.

Q5. Reliable Data Transfer: Timeout and Delayed ACK

A reliable data transfer protocol uses alternating sequence numbers 0 and 1. The sender maintains a timer for the currently outstanding packet and retransmits the packet when the timer expires.

The following events occur:

Time	Event
t_0	Sender transmits DATA_0
t_1	Receiver receives DATA_0
t_2	Receiver sends ACK_0
t_3	Sender timeout occurs
t_4	Sender retransmits DATA_0
t_5	Original ACK_0 arrives at the sender
t_6	Retransmitted DATA_0 arrives at the receiver

Assume that the timeout at t_3 occurs because the ACK is delayed rather than lost. Also assume that the receiver delivers each correctly received sequence number to the application at most once.

(a) Explain why the sender retransmits DATA_0 even though the receiver has already received it.

(b) What should the receiver do when the retransmitted DATA_0 arrives at t_6?

(c) What should the sender do when ACK_0 arrives at t_5?

(d) Should the receiver send another ACK_0 after detecting the duplicate DATA_0? Justify your answer.

(e) Explain how the use of sequence numbers prevents the duplicate DATA_0 from being delivered twice to the application.

(f) What problem would occur if the protocol did not use sequence numbers?

Q6. Go-Back-N: Window Size and Throughput

A Go-Back-N protocol operates over a link with the following parameters:

$$R = 10 \text{ Mbps}, \qquad L = 1000 \text{ bytes}, \qquad RTT = 80 \text{ ms}.$$

The sender window size is $N = 4$. Assume:

- ACK transmission time is negligible,
- processing and queuing delays are negligible,
- $RTT = 80$ ms includes the propagation delay and ACK-related delay,
- there are no packet losses,
- the sender follows standard Go-Back-N sliding-window behavior.

For throughput calculations, use the standard window-limited approximation

$$T = \min\left(R, \frac{NL}{RTT}\right),$$

where T is the achievable throughput, R is the link rate, N is the sender window size, L is the packet size in bits, and RTT is the round-trip time.

(a) Calculate the transmission time of one 1000-byte packet.

(b) Calculate the maximum amount of unacknowledged data that can be outstanding with a sender window size of 4.

(c) Determine whether the sender can continuously transmit packets at the full link rate. Justify your answer quantitatively.

(d) Calculate the maximum achievable throughput imposed by the sender window size using the given throughput approximation.

(e) Determine the minimum integer window size required to fully utilize the 10 Mbps link using the given throughput approximation.

(f) Explain qualitatively why packet losses can reduce the throughput of Go-Back-N more severely than a protocol that retransmits only the lost packet.